5

# Supervised Learning

# التعلم بمشرف

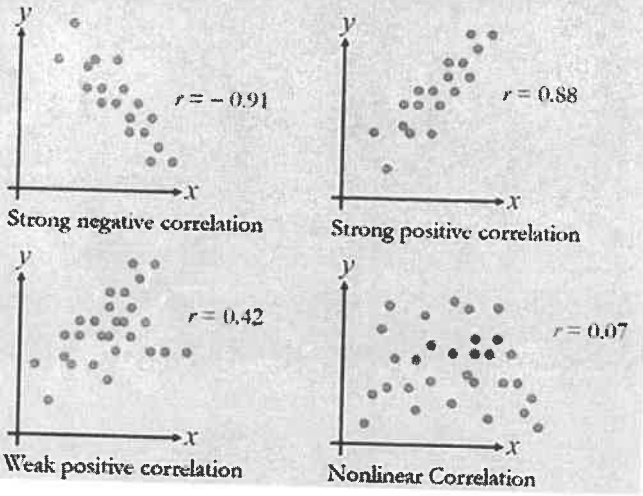
| Input (X) | Output (Y) | Application |
|---|---|---|
| email → | spam? (0/1) | spam filtering |
| audio → | text transcripts | speech recognition |
| English → | Spanish | machine translation |
| ad, user info → | click? (0/1) | online advertising |
| image, radar → | position of other cars | self-driving car |

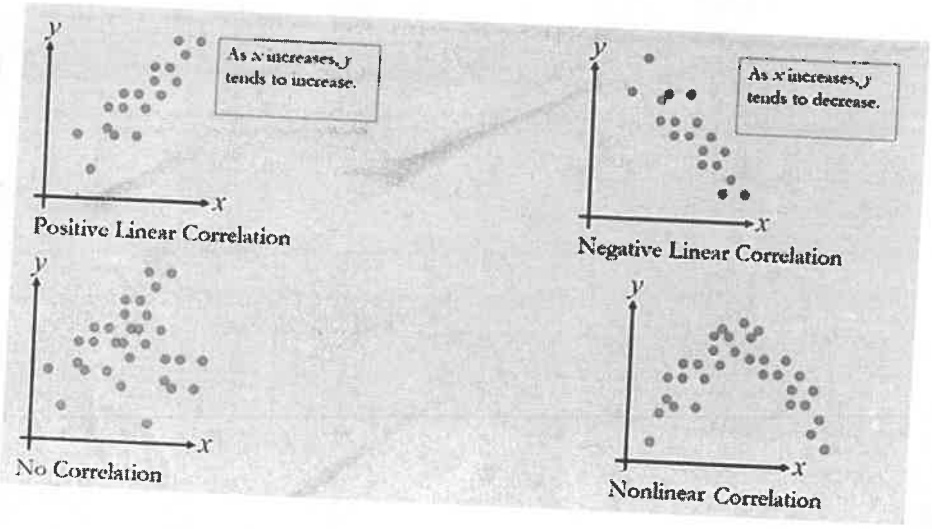
## Correlation

- A correlation is a relationship between two variables. The data can be represented by the ordered pairs (x, y) where x is the independent variable, and y is the dependent variable.
- The **independent variable** is the **cause**. Its value is *independent* of other variables in your study.
- The **dependent variable** is the **effect**. Its value *depends* on changes in the independent variable.
- A scatter plot can be used to determine whether a linear (straight line) correlation exists between two variables.

## Correlation Coefficient

- $-1 \le r \le +1$
- If x and y have a strong positive linear correlation, r is close to 1.
- If x and y have a strong negative linear correlation, r is close to −1.
- If there is no linear correlation or a weak linear correlation, r is close to 0.






## Correlation Coefficient

- The **correlation coefficient** is a measure of the strength and the direction of a linear relationship between two variables. The symbol r represents the sample correlation coefficient. The formula for r is
- The range of the correlation coefficient is **–1 to 1.**

1. **List your data points** for variables $x$ and $y$.
2. **Calculate the sums** $(\sum x, \sum y, \sum xy, \sum x^2, \sum y^2)$.
3. **Calculate the number of data points** $(n)$.
4. **Plug the sums into the formula:**

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}.$$

**Example:**

Calculate the correlation coefficient $r$ for the following data.

| $x$ | $y$ | $xy$ | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 1 | −3 | −3 | 1 | 9 |
| 2 | −1 | −2 | 4 | 1 |
| 3 | 0 | 0 | 9 | 0 |
| 4 | 1 | 4 | 16 | 1 |
| 5 | 2 | 10 | 25 | 4 |
| $\sum x = 15$ | $\sum y = -1$ | $\sum xy = 9$ | $\sum x^2 = 55$ | $\sum y^2 = 15$ |

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}} = \frac{5(9) - (15)(-1)}{\sqrt{5(55) - 15^2}\sqrt{5(15) - (-1)^2}}$$

ما هي تطبيقات صيغة معامل الارتباط؟

فيما يلي أهم تطبيقات صيغة معامل الارتباط:

- تساعد صيغة معامل الارتباط في تحليل البيانات المقدمة من خلال تحديد الدرجة التي ترتبط بها متغيرين مما يصور بشكل أكبر العلاقة الخطية بين متغيرين.
- يتم استخدامه للتحليل المالي لأنه يحدد العلاقة بين مجموعات البيانات في الأعمال التجارية وبالتالي يدعم عملية اتخاذ القرار.
- فهو يساعد كثيراً في اتخاذ القرارات في مختلف المجالات لأنه يساعد على فهم قوة العلاقة بين متغيرين مختلفين.

**Example 1:** Calculate the correlation coefficient of the given data.

| x | 50 | 51 | 52 | 53 | 54 |
|---|---|---|---|---|---|
| y | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 |

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}$$

**Solution:**

Here, n = 5

| x | 50 | 51 | 52 | 53 | 54 |
|---|---|---|---|---|---|
| y | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 |
| xy | 155 | 163.2 | 171.6 | 180.2 | 189 |
| $x^2$ | 2500 | 2601 | 2704 | 2809 | 2916 |
| $y^2$ | 9.61 | 10.24 | 10.89 | 11.56 | 12.25 |

$\sum x = 260$

$\sum y = 16.5$

$\sum xy = 859$

$\sum x^2 = 13530$

$\sum y^2 = 54.55$

By substituting all the values in the formula, we get r = 1. This shows a positive correlation coefficient.

If x & y are the two variables of discussion, then the correlation coefficient can be calculated using the formula

## Example 1

Calculate the correlation coefficient $r$ for the following data.

| $x$ | $y$ | $xy$ | $x^2$ | $y^2$ |
|---|---|---|---|---|
| 1 | −3 | −3 | 1 | 9 |
| 2 | −1 | −2 | 4 | 1 |
| 3 | 0 | 0 | 9 | 0 |
| 4 | 1 | 4 | 16 | 1 |
| 5 | 2 | 10 | 25 | 4 |
| $\Sigma x = 15$ | $\Sigma y = -1$ | $\Sigma xy = 9$ | $\Sigma x^2 = 55$ | $\Sigma y^2 = 15$ |

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}} = \frac{5(9) - (15)(-1)}{\sqrt{5(55) - 15^2}\sqrt{5(15) - (-1)^2}} \approx 0.986$$

There is a strong positive linear correlation between $x$ and $y$.

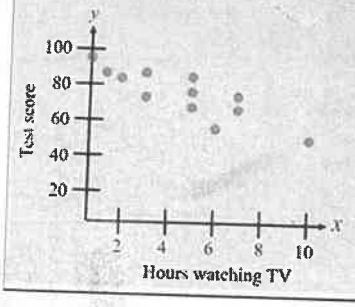
## Example 2

The following data represents the number of hours 12 different students watched television during the weekend and the scores of each student who took a test the following Monday.

Calculate the correlation coefficient $r$.

Strong negative | Week negative | Week positive | Strong positive

-1 -0.5 0 0.5 1




| Hours, $x$ | 0 | 1 | 2 | 3 | 3 | 5 | 5 | 5 | 6 | 7 | 7 | 10 | $\Sigma x = 54$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Test score, $y$ | 96 | 85 | 82 | 74 | 95 | 68 | 76 | 84 | 58 | 65 | 75 | 50 | $\Sigma y = 908$ |
| $xy$ | 0 | 85 | 164 | 222 | 285 | 340 | 380 | 420 | 348 | 455 | 525 | 500 | $\Sigma xy = 3724$ |
| $x^2$ | 0 | 1 | 4 | 9 | 9 | 25 | 25 | 25 | 36 | 49 | 49 | 100 | $\Sigma x^2 = 332$ |
| $y^2$ | 9216 | 7225 | 6724 | 5476 | 9025 | 4624 | 5776 | 7056 | 3364 | 4225 | 5625 | 2500 | $\Sigma y^2 = 70836$ |

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}} = \frac{12(3724) - (54)(908)}{\sqrt{12(332) - 54^2}\sqrt{12(70836) - (908)^2}} \approx -0.831$$

- There is a strong negative linear correlation.
- As the number of hours spent watching TV increases, the test scores tend to decrease

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- A labeled training sample is a collection of instances $(x_1, y_1) \ldots, (x_n, y_n)$
- Assume $(x_i, y_i) \overset{i.i.d.}{\sim} P(x,y)$. Again, $P(x,y)$ is unknown
- Supervised learning learns a function $f: X \rightarrow Y$ in some function family $F$, such that $f(x)$ predicts the true label y on future data $x$, where $(x,y) \overset{i.i.d.}{\sim} P(x,y)$
  - Classification: if y discrete
  - Regression: if y continuous

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MSE = (1/N) * Σ (y_i - ŷ_i)^2
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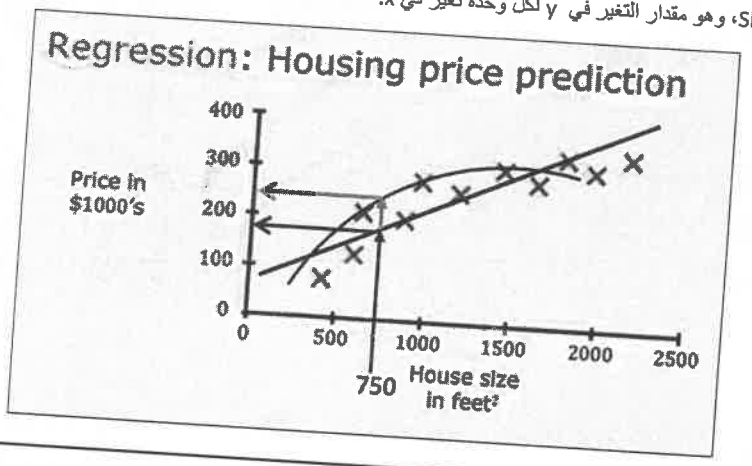
معادلة الانحدار الخطي البسيط هي معادلة رياضية تصف العلاقة الخطية بين متغيرين: متغير تابع (y) ومتغير مستقل (x) .الصيغة الأساسية هي $y = b + mx$، حيث تمثل b نقطة التقاطع مع المحور y القيمة المتوقعة لـ y عندما يكون $x = 0$ وتمثل m ميل الخط الذي يصف معدل التغير في y لكل وحدة تغير في x. وتهدف إلى العثور على المعلمات التي تناسب بشكل أفضل مجموعة معينة من نقاط البيانات.

- y المتغير التابع أو المستجيب.
- x لمتغير المستقل أو المتنبئ.
- b نقطة التقاطع (Intercept)، وهي قيمة yعندما تكون x تساوي صفرًا.
- m ميل الخط ((Slope، وهو مقدار التغير في y لكل وحدة تغير في x.




## Regression Line

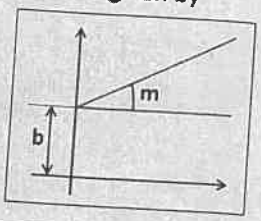
- **A regression line**, also called a line of best fit, is the line for which the sum of the squares of the residuals is a minimum.
- **The equation of a regression line** for an independent variable x and a dependent variable y is

$$\hat{y} = mx + b$$

- where $\hat{y}$ is the predicted y-value for a given x-value. The slope m and y-intercept b are given by

$$m = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2} \text{ and } b = \bar{y} - m\bar{x} = \frac{\sum y}{n} - m\frac{\sum x}{n}$$

- The regression line always passes through $(\bar{x}, \bar{y})$.



## Solution

| Hours, $x$ | 0 | 1 | 2 | 3 | 3 | 5 | 5 | 5 | 6 | 7 | 7 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Test score, $y$ | 96 | 85 | 82 | 74 | 95 | 68 | 76 | 84 | 58 | 65 | 75 | 50 |
| $xy$ | 0 | 85 | 164 | 222 | 285 | 340 | 380 | 420 | 348 | 455 | 525 | 500 |
| $x^2$ | 0 | 1 | 4 | 9 | 9 | 25 | 25 | 25 | 36 | 49 | 49 | 100 |
| $y^2$ | 9216 | 7225 | 6724 | 5476 | 9025 | 4624 | 5776 | 7056 | 3364 | 4225 | 5625 | 2500 |

$\sum x = 54$
$\sum y = 908$
$\sum xy = 3724$
$\sum x^2 = 332$
$\sum y^2 = 70836$

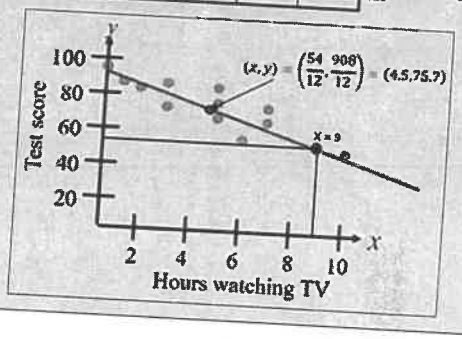
$$m = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2} = \frac{12(3724) - (54)(908)}{12(332) - (54)^2} \approx -4.067$$

$$b = \bar{y} - m\bar{x} = \frac{908}{12} - (-4.067)\frac{54}{12} \approx 93.97$$

$\hat{y} = -4.07x + 93.97$

$\hat{y} = -4.07\,(9) + 93.97 = 57.34$

A student who watches 9 hours of TV over the weekend can expect to receive about a 57.34 on Monday's test.




# Evaluation

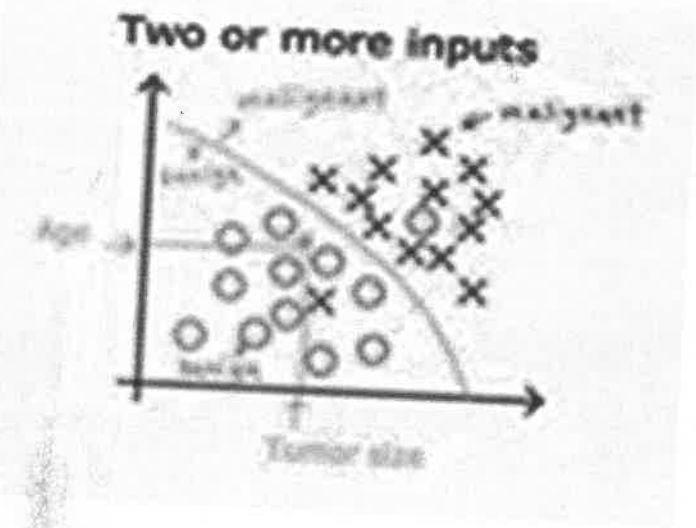
Training set error

– 0-1 loss for classification $\frac{1}{n}\sum_{i=1}^{n}(f(x_i) \neq \hat{y}_i)$,

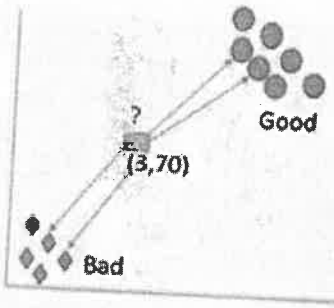
**Root Mean Squared Error (RMSE) =** $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

**where:**

- N is the number of data points in the training set.
- $y_i$ is the actual value for the i-th data point .
- $\hat{y}_i$ is the predicted value for the i-th data point.



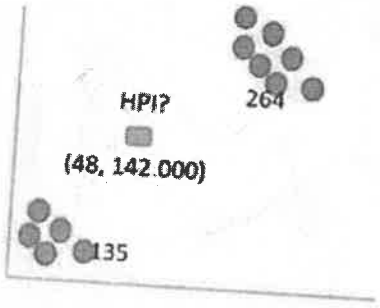

K=4




| Name | Cigarettes | Weight | Heart Attack | Distance | |
|---|---|---|---|---|---|
| A | 7 | 70 | Bad | $\sqrt{(3-7)^2 + +(70-70)^2} = 4$ | 1 |
| B | 7 | 40 | Bad | $\sqrt{(3-7)^2 + +(70-40)^2} = 30.27$ | 4 |
| C | 3 | 40 | Good | $\sqrt{(3-3)^2 + +(70-40)^2} = 30.00$ | 2 |
| D | 1 | 40 | Good | $\sqrt{(3-1)^2 + +(70-40)^2} = 30,07$ | 3 |
| E | 3 | 70 | Bad | | |

# Regression Using KNN

قيم عددية




| Age | Loan | House Price Index | Distance | |
|---|---|---|---|---|
| 25 | $40,000 | 135 | 102000 | |
| 35 | $60,000 | 256 | 82000 | |
| 45 | $80,000 | 231 | 62000 | |
| 20 | $20,000 | 267 | 122000 | |
| 35 | $120,000 | 139 | 22000 | 2 |
| 52 | $18,000 | 150 | 124000 | |
| 23 | $95,000 | 127 | 47000 | |
| 40 | $62,000 | 216 | 80000 | |
| 60 | $100,000 | 139 | 42000 | 3 |
| 48 | $220,000 | 250 | 78000 | |
| 33 | $150,000 | 264 | 8000 | 1 |
| 48 | $142,000 | ? | | |

$$D=\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$$

# Explanation

K=1

- using the training set to classify an unknown case
- (Age=33 and Loan=$150,000) (Euclidean distance).
- If K=1 then the nearest neighbor is the last case in the set
- with HPI=264.
- D = Sqrt[(48-33)^2 + (142000-150000)^2] = 8000.01 >> HPI = 264

K=3

- the prediction for HPI is equal to the average of HPI for the top three neighbors
- HPI = (264+139+139)/3 = 180.7

TRY when K =4? What is HPI for it ?

| Age | Loan | House Price Index | Distance | |
|---|---|---|---|---|
| 25 | $40,000 | 135 | 102000 | |
| 35 | $60,000 | 256 | 82000 | |
| 45 | $80,000 | 231 | 62000 | |
| 20 | $20,000 | 267 | 122000 | |
| 35 | $120,000 | 139 | 22000 | 2 |
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| 48 | $142,000 | ? | | |

$$D=\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$$